GATE ME 90 Days Challenge

Day 79/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Full Syllabus	Full-Length Mock 3	19 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on a mixed full-syllabus warm-up for the third full-length mock: word usage, percentage change, determinants, integration by parts, concurrent forces, hollow shafts, Kutzbach criterion, cantilever vibration, stress concentration, centre of pressure, heat-exchanger arrangements, compressor energy balance, summer air conditioning, bend allowance and moving-average forecasting. Try them in **60 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE. **Today's test:** open 7 to 11 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 65 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

Select the word that correctly completes the sentence: The new heat-treatment schedule had a marked _____ on the surface hardness of the gears.

(A) affect
(B) effect
(C) affection
(D) effectuate

Q2 NAT

The price of diesel rises by 25 %. By what percentage must a transport firm reduce its diesel consumption so that its spending on diesel stays the same? _____

Q3 MCQ

A is a 3 × 3 matrix with det(A) = 4. The value of $\det(2A^{-1})$ is

(A) 1/2
(B) 8
(C) 32
(D) 2

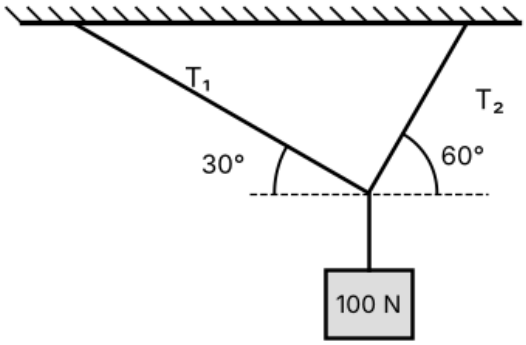
Q4 MCQ

The value of the integral of $x\,e^{x}$ from x = 0 to x = 1 is

(A) 1
(B) e − 1
(C) e
(D) 1/2

Q5 NAT

A weight of 100 N hangs from two light strings tied to a ceiling, as shown. One string makes 30° and the other 60° with the horizontal. The tension in the string inclined at 60° (in N, round off to one decimal place) is _____



Q6 NAT

A hollow steel shaft has an outer diameter of 80 mm and an inner diameter of 60 mm. It transmits a torque of 5 kN·m. The maximum shear stress in the shaft (in MPa, round off to one decimal place) is _____

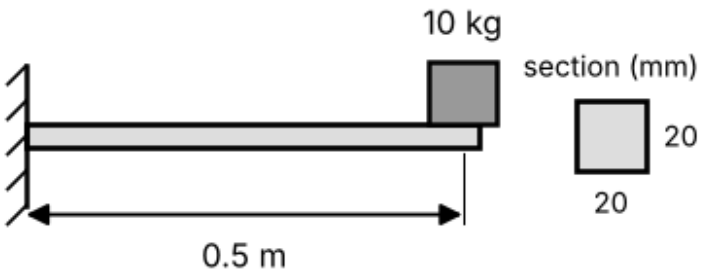
Q7 MCQ

A planar mechanism has five links (including the fixed link) joined in a single closed loop by five revolute (turning) pairs. Its degree of freedom is

(A) 1
(B) 0
(C) 3
(D) 2

Q8 NAT

A light steel cantilever (E = 200 GPa) of length 0.5 m and square cross-section 20 mm × 20 mm carries a mass of 10 kg at its free end, as shown. Neglecting the mass of the beam, the natural frequency of transverse vibration (in Hz, round off to two decimal places) is _____



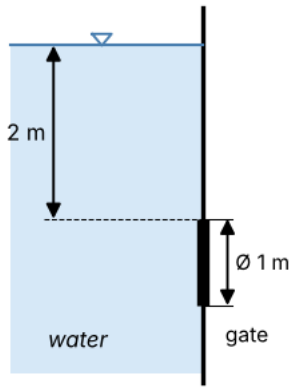
Q9 MCQ

For a component of a *ductile* material carrying a *static* load, the theoretical stress concentration factor at a notch is usually ignored in design because

(A) the notch raises the yield strength of the material
(B) local yielding at the notch root redistributes the stress over the section
(C) ductile materials are not sensitive to notches under any loading
(D) the factor applies only to compressive loads

Q10 NAT

A vertical circular gate of 1 m diameter closes an opening in the wall of a water tank, as shown. Its top edge is 2 m below the free surface. The depth of the centre of pressure below the free surface (in m, round off to three decimal places) is _____



Q11 MCQ

Which one of the following statements about heat exchangers is correct?

(A) In a counter-flow exchanger, the outlet temperature of the cold fluid can be higher than the outlet temperature of the hot fluid.

(B) In a parallel-flow exchanger, the outlet temperature of the cold fluid can be higher than the outlet temperature of the hot fluid.

(C) For the same inlet and outlet temperatures, the LMTD of a parallel-flow exchanger exceeds that of a counter-flow exchanger.

(D) When one fluid condenses at constant temperature, a counter-flow arrangement gives a higher effectiveness than a parallel-flow one.

Q12 NAT

Air (c_p = 1.005 kJ/kg·K) flows at 1 kg/s through an adiabatic compressor and its temperature rises from 300 K to 450 K. Neglecting changes in kinetic and potential energy, the power required (in kW, round off to one decimal place) is _____

Q13 MCQ

In a summer air-conditioning system for a humid climate, the usual sequence of processes applied to the supply air is

(A) heating followed by humidification

(B) adiabatic (evaporative) cooling only

(C) cooling and dehumidification followed by reheating

(D) sensible cooling followed by humidification

Q14 NAT

A 2 mm thick sheet is bent through 90° with an inside bend radius of 5 mm. Taking the neutral axis at 0.4 times the thickness from the inside surface, the bend allowance (developed length of the bend, in mm, round off to two decimal places) is _____

Q15 MCQ

The monthly demand of an item from January to June was 120, 132, 126, 138, 144 and 129 units. The three-month simple moving-average forecast for July is

(A) 129 units

(B) 136 units

(C) 131.5 units

(D) 137 units

Answer Key

Q	Type	Answer
Q1	MCQ	(B) effect
Q2	NAT	20
Q3	MCQ	(D) 2
Q4	MCQ	(A) 1
Q5	NAT	86.6 (accept 85.8 to 87.4)
Q6	NAT	72.8 (accept 72.4 to 73.2)
Q7	MCQ	(D) 2
Q8	NAT	12.73 (accept 12.60 to 12.86)
Q9	MCQ	(B) local yielding at the notch root redistributes the stress over the section
Q10	NAT	2.525 (accept 2.510 to 2.540)
Q11	MCQ	(A) In a counter-flow exchanger, the outlet temperature of the cold fluid can be higher than the outlet temperature of the hot fluid.
Q12	NAT	150.8 (accept 149.3 to 152.3)
Q13	MCQ	(C) cooling and dehumidification followed by reheating
Q14	NAT	9.11 (accept 9.02 to 9.20)
Q15	MCQ	(D) 137 units

Detailed Solutions

Q1 **Answer: (B) effect**

Concept: 'Effect' is usually a noun (a result); 'affect' is usually a verb (to influence).

The blank follows the article 'a marked ...', so a noun is needed.

The noun meaning 'result' is **effect**, option (B).

Exam tip: Remember 'A for Action' (affect is the verb) and 'E for End result' (effect is the noun).

Q2 **Answer: 20**

Concept: Spending = price × quantity; if spending is fixed, the quantity changes by the inverse ratio of the price.

New price = 1.25 × old, so new quantity = old/1.25 = 0.8 × old.

Reduction = 1 − 0.8 = **20 %**.

Exam tip: For an increase of r %, the cut needed is r/(100 + r) × 100 % — not r %. Here 25/125 × 100 = 20 %.

Q3 **Answer: (D) 2**

Concept: For an n × n matrix, $\det(kA) = k^n \det(A)$ and $\det(A^{-1}) = 1/\det(A)$.

$\det(2A^{-1}) = 2^3 \times \det(A^{-1}) = 8 \times (1/4)$.

= **2**, option (D).

Exam tip: The scalar comes out raised to the order of the matrix — 1/2 comes from forgetting the cube.

Q4 **Answer: (A) 1**

Concept: Integration by parts: $\int u\,dv = uv - \int v\,du$, with u = x and $dv = e^x\,dx$.

$\int x\,e^x\,dx = x\,e^x - e^x = (x - 1)e^x$.

From 0 to 1: $(0)e - (-1)(1) =$ **1**, option (A).

Exam tip: Differentiate $(x - 1)e^x$ to check: you get $x\,e^x$.

Q5 **Answer: 86.6 (accept 85.8 to 87.4)**

Concept: Three concurrent forces in equilibrium — use Lami's theorem: each force is proportional to the sine of the angle between the other two.

Angles between the forces: between the strings 90°; between the weight and the 60° string 150°; between the weight and the 30° string 120°. Lami: $T_2/\sin 120° = W/\sin 90°$, so $T_2 = 100 \times 0.866$.

Check by resolving:

Horizontal: $T_1 \cos 30° = T_2 \cos 60° \rightarrow T_1 = 0.577\,T_2$. Vertical: $T_1 \sin 30° + T_2 \sin 60° = 100 \rightarrow 1.155\,T_2 = 100$.

T_2 (60° string) = **86.6 N**, and T_1 = 50.0 N.

Exam tip: The steeper string always carries the larger share of a hanging load.

Q6 **Answer: 72.8 (accept 72.4 to 73.2)**

Concept: $\tau_{max} = T\,r_o/J$ with $J = \pi(D^4 - d^4)/32$ for a hollow shaft.

$J = \pi(80^4 - 60^4)/32 = \pi \times 2.80 \times 10^7/32 = 2.749 \times 10^6\ mm^4$.

$\tau_{max} = 5 \times 10^6 \times 40/2.749 \times 10^6 =$ **72.8 MPa** (at the outer surface).

Exam tip: The stress at the inner surface is 72.8 × 30/40 = 54.6 MPa — never zero in a hollow shaft.

Q7 **Answer: (D) 2**

Concept: Kutzbach criterion for planar mechanisms: $F = 3(n - 1) - 2j - h$.

n = 5, j = 5 (lower pairs), h = 0.

$F = 3 \times 4 - 2 \times 5 =$ **2**, option (D): two independent inputs are needed.

Exam tip: Inserting one more link and one more pin into a closed loop adds 3 − 2 = 1 degree of freedom, so a five-bar chain has one more DOF than a four-bar chain.

Q8 **Answer: 12.73 (accept 12.60 to 12.86)**

Concept: The end stiffness of a cantilever is $k = 3EI/L^3$; then $f_n = (1/2\pi)\sqrt{(k/m)}$.

$I = 20 \times 20^3/12 = 13\,333\ mm^4$. $k = 3 \times 200\,000 \times 13\,333/500^3 = 64\ N/mm = 64\,000\ N/m$.

$f_n = (1/2\pi)\sqrt{(64\,000/10)} = 80/(2\pi) =$ **12.73 Hz**.

Exam tip: Equivalently $f_n = (1/2\pi)\sqrt{(g/\delta_{st})}$ with $\delta_{st} = mg/k = 1.53$ mm — a quick cross-check.

Q9 **Answer: (B) local yielding at the notch root redistributes the stress over the section**

Concept: A stress concentration factor gives the elastic peak stress at a notch.

In a ductile material the small region at the notch root yields first, and the load is shared by the surrounding material, so the peak is relieved before failure.

Hence for static loads on ductile parts the factor is usually ignored: **(B)**. For brittle materials, and for fatigue loading of any material, it must be applied.

Exam tip: Option (C) is the trap — ductile parts are very notch-sensitive in fatigue.

Q10 **Answer: 2.525 (accept 2.510 to 2.540)**

Concept: $h_{cp} = \bar{h} + I_G/(A\,\bar{h})$ for a vertical plane surface.

$\bar{h} = 2 + 0.5 = 2.5$ m; $A = \pi/4 \times 1^2 = 0.7854\ m^2$; $I_G = \pi \times 1^4/64 = 0.04909\ m^4$.

$h_{cp} = 2.5 + 0.04909/(0.7854 \times 2.5) = 2.5 + 0.025 =$ **2.525 m**. (The resultant force is $\rho g\,\bar{h}\,A = 19.3$ kN.)

Exam tip: For a circle $I_G/A = d^2/16$, so $h_{cp} - \bar{h} = d^2/(16\bar{h})$ — no need to compute I_G separately.

Q11 **Answer: (A) In a counter-flow exchanger, the outlet temperature of the cold fluid can be higher than the outlet temperature of the hot fluid.**

Concept: In counter-flow the cold fluid leaves next to the hot inlet; in parallel flow both leave at the same end.

So in counter-flow $T_{c,out}$ may exceed $T_{h,out}$ (it is limited only by $T_{h,in}$), while in parallel flow $T_{c,out}$ can at most approach $T_{h,out}$.

Counter-flow has the larger LMTD, and with a condensing fluid ($C_r = 0$) both arrangements give $\varepsilon = 1 - e^{-NTU}$. Answer: **(A)**.

Exam tip: 'Temperature cross' is possible only in counter-flow.

Q12 **Answer: 150.8 (accept 149.3 to 152.3)**

Concept: Steady-flow energy equation for an adiabatic device: $W = \dot{m}(h_2 - h_1) = \dot{m}c_p(T_2 - T_1)$.

$W = 1 \times 1.005 \times (450 - 300)$.

$=$ **150.8 kW** (input).

Exam tip: Use c_p (enthalpy) for flow devices; c_v would give 107.7 kW, a common slip.

Q13 **Answer: (C) cooling and dehumidification followed by reheating**

Concept: Hot humid air must lose both sensible heat and moisture.

The cooling coil is run below the dew point so that moisture condenses (cooling and dehumidification); the air then often becomes too cold and is reheated to the required supply temperature.

Answer: **(C)**.

Exam tip: Evaporative cooling (B) adds moisture — it suits hot *dry* climates only.

Q14 **Answer: 9.11 (accept 9.02 to 9.20)**

Concept: Bend allowance = $\alpha(R + kt)$, the arc length of the neutral axis, with α in radians.

$R + kt = 5 + 0.4 \times 2 = 5.8$ mm; $\alpha = \pi/2$.

Bend allowance = $1.5708 \times 5.8 =$ **9.11 mm**.

Exam tip: For small R/t the neutral axis shifts towards the inside ($k \approx 0.33$–0.4); for large R/t it is near mid-thickness ($k = 0.5$).

Q15 **Answer: (D) 137 units**

Concept: An n-period moving average forecasts the next period as the mean of the latest n actual values.

Latest three months: April 138, May 144, June 129.

Forecast = $(138 + 144 + 129)/3 = 411/3 =$ **137 units**, option (D).

Exam tip: The mean of all six months is 131.5 — use only the last n values.

What next?

Day 79 test tonight, 7 to 11 PM

65 questions · 180 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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